

# LINUX ADMINISTRATION

## LAB 1:

INITIALISATION

DHCP: ISC KEA DHCP + HA HOOK

DNS: BIND9



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Hey there! ☒

I'm **Hosni Zaaraoui**, a passionate Linux enthusiast on a mission to become the best version of myself as a Linux & Network Administrator.

This series isn't just a dry set of commands — it's my personal journey of building, breaking, securing, and experimenting with real-world IT infrastructure in a lab environment.

If you've ever wanted to configure Linux services from scratch, secure them like a pro, and test them in a safe network — you're in the right place.

We'll be doing things like:

- Installing & configuring core services (on Ubuntu 22.04)
- Setting up a firewall with pfSense
- Playing with FreeNAS for storage
- Testing security with Kali Linux

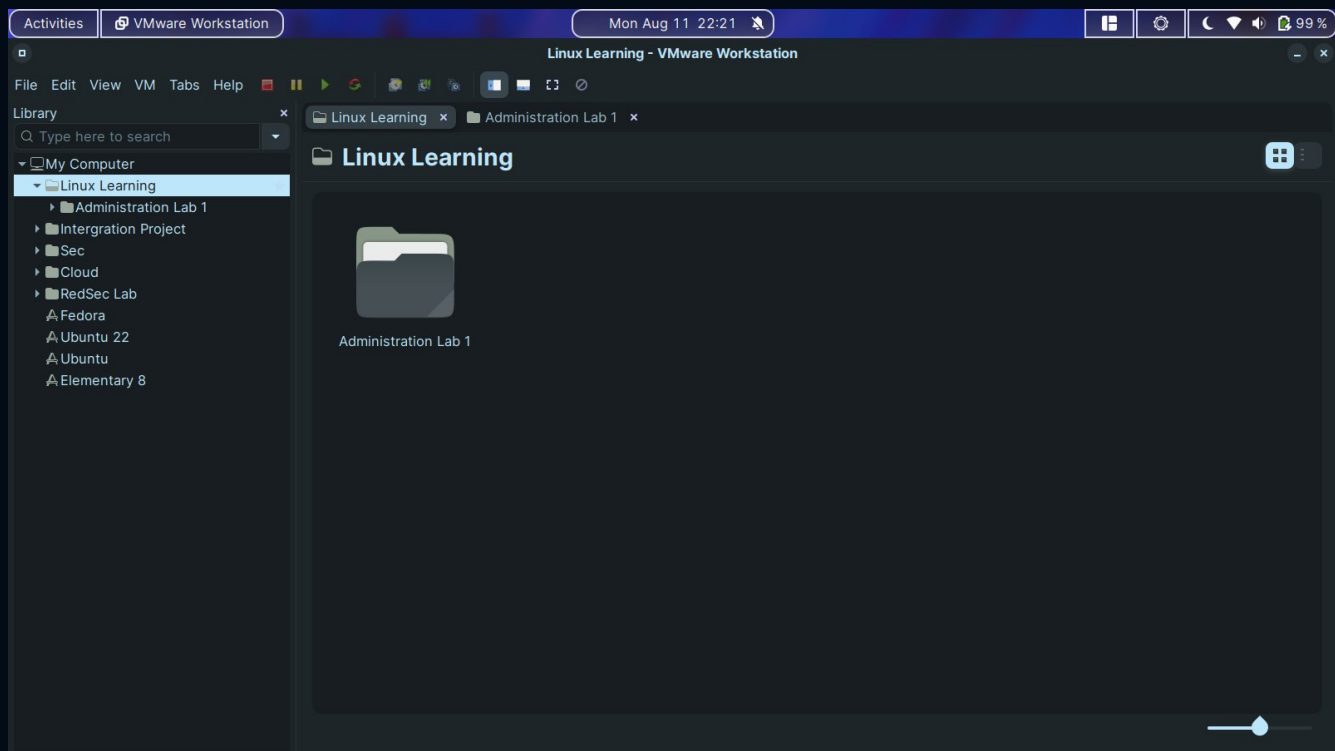
And yes — you'll see every single step, with explanations that actually make sense.

# KEEPING THINGS ORGANIZED

I like my labs to be neat and easy to manage, so for every new lab I create, I make a dedicated folder in VMware Workstation.

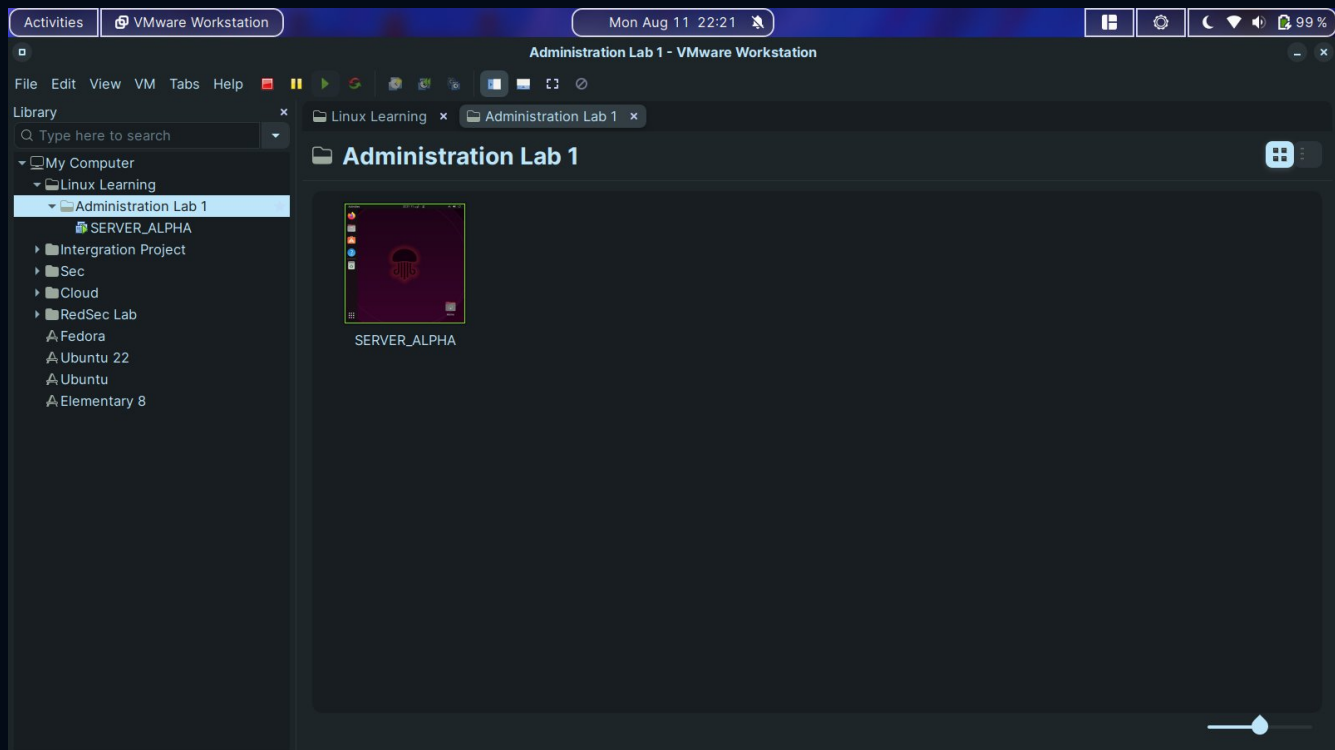
For example:

## Linux Learning/Administration Lab 1



# SETTING UP OUR FIRST LAB VMs

For this lab, I'll be using two Ubuntu 22.04 servers and a windows client.



Here's my workflow:

1. Start with one VM and install all the packages/services we need.
2. Once it's set up, clone it to create the second VM — saving time and ensuring consistency.

To keep our lab isolated and realistic:

- I'm using a dedicated **Host-Only network (VMnet) with DHCP disabled**.
- I'm adding a second **NAT adapter** so the VMs can still access the internet for package installation.

# EPISODE 1: INITIALISATION

Let's get our Linux server ready for action.

We start here because a solid foundation will make everything else smoother.

## Step 1 – Update Your System (Always!)

The very first command I run on any fresh install is:

```
sudo apt update && sudo apt upgrade -y
```

**Why?** This makes sure you have the latest security patches and software improvements. Think of it as giving your server a fresh cup of coffee before starting work!

## Step 2 – Install VMware Tools (For Virtual Machines)

If you're running this on VMware, installing VMware Tools makes everything smoother — better performance, shared clipboard, and easier file transfers.

```
sudo apt install open-vm-tools-desktop
```

### Step 3 – Give Your Server a Static IP

Servers need a fixed address so you can always find them on the network.

We'll use Netplan to set that up.

Open the Netplan config file:

```
sudo nano /etc/netplan/01-network-manager-all.yaml
```

Add the ethernets section:

```
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens33:
      addresses:
        - 192.168.50.10/24
```

**Tip:** In nano, press ALT+M to enable mouse support — it makes navigation much easier :)

### Step 4 – Save & Apply

In nano, save with CTRL+S, exit with CTRL+X, then apply the config:

```
sudo netplan apply
```

Check your shiny new IP:

```
ip a
```

**And that's it!**

You've just done the most important part — your server is updated, optimized, and has a reliable address on the network.

With this done, we're ready to start building real capabilities in the next episode.

## EPISODE 2 : SETTING UP DHCP SERVICE

It's time to give our server the power to automatically hand out IP addresses using KEA DHCP, a modern and high-performance DHCP server from ISC.

### 1. Add the KEA repository

We start by pulling the official KEA repository setup script and running it:

```
curl -1sLf 'https://dl.cloudsmith.io/public/isc/kea-3-0/setup.deb.sh' | sudo -E bash
```

### 2. Install KEA DHCP packages

Next, we install the main DHCPv4 server, the admin tools, and the hooks framework (which allows adding extra features):

```
sudo apt install -y isc-kea-dhcp4 isc-kea-admin isc-kea-hooks
```

### 3. Prepare the configuration

Before creating our own configuration, we'll move the default one to a backup file so we can restore it if needed:

```
sudo mv /etc/kea/kea-dhcp4.conf /etc/kea/kea-dhcp4.conf.bkp
```

## 4. Open the configuration file

Now we'll create and edit a fresh configuration file.

You can use any text editor you like, but here I'll use nano:

```
sudo nano /etc/kea/kea-dhcp4.conf
```

In the next section, we will write the complete configuration from scratch and explain every single line so you know exactly what each part does.

Below is our `kea-dhcp4.conf` file.

We'll go through it step-by-step so you know exactly what each setting means.

```
{
  "Dhcp4": {
    // Listen for DHCP requests on the ens33 interface
    "interfaces-config": {
      "interfaces": ["ens33"]
    },
    // Control socket for runtime commands (without restarting the service)
    "control-socket": {
      "socket-type": "unix",
      "socket-name": "kea4-ctrl-socket"
    },
  },
}
```

```
// Store lease data in memory and persist it to disk
```

```
"lease-database": {  
  "type": "memfile",  
  "persist": true  
},
```

```
// IPv4 subnet configuration
```

```
"subnet4": [  
  {  
    "id": 1, // Unique ID for this subnet  
    "subnet": "192.168.50.0/24", // Network address and  
mask  
    "pools": [  
      {  
        "pool": "192.168.50.100 - 192.168.50.150" // Range  
of assignable IPs  
      }  
    ],  
    "option-data": [  
      {  
        "name": "routers",  
        "data": "192.168.50.1" // Default gateway  
      },  
      {  
        "name": "domain-name-servers",
```

```

        "data": "192.168.50.10,192.168.50.11" // DNS
servers
    }
]
},
],

// Load the High Availability (HA) hook library
"hooks-libraries": [
    {
        "library": "/usr/lib/x86_64-linux-
gnu/kea/hooks/libdhcp_ha.so",
        "parameters": {
            "high-availability": [
                {
                    "this-server-name": "server-alpha", // Name of
this DHCP server
                    "mode": "hot-standby", // HA mode
                    "auto-failover": true, // Automatically failover
to standby
                    "heartbeat-delay": 10000, // Time between health
checks (ms)
                    "max-response-delay": 60000, // Max wait before
failover (ms)
                    "max-unacked-clients": 0, // No limit for
unacknowledged clients
                    "peers": [

```

```
        {
            "name": "server-alpha",
            "url": "http://192.168.50.10:8000/",
            "role": "primary"
        },
        {
            "name": "server-beta",
            "url": "http://192.168.50.11:8000/",
            "role": "standby"
        }
    ]
}

]
}

},

],
```

*// Logging configuration*

```
"loggers": [
    {
        "name": "kea-dhcp4", // Logger name
        "output-options": [
            {
```

```
        "output": "/var/log/kea/kea-dhcp4.log", // Log file
        path
        "pattern": "%-5p %m\n", // Log format
        "flush": false,
        "maxsize": 1048576, // Max log size in bytes
        "maxver": 8 // Number of rotated log files to keep
    }
],
    "severity": "INFO", // Logging level
    "debuglevel": 0 // Debug level (0 = none)
}
]
}
}
```

## 5. Testing Our DHCP Server

After saving the configuration in nano (CTRL+S then CTRL+X), we test it to ensure there are no syntax errors:

```
sudo kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
```

If the test passes with no errors, start the service:

```
sudo systemctl start isc-kea-dhcp4-server
```

Enable it so it starts automatically on boot:

```
sudo systemctl enable isc-kea-dhcp4-server
```

## Checking with a Client

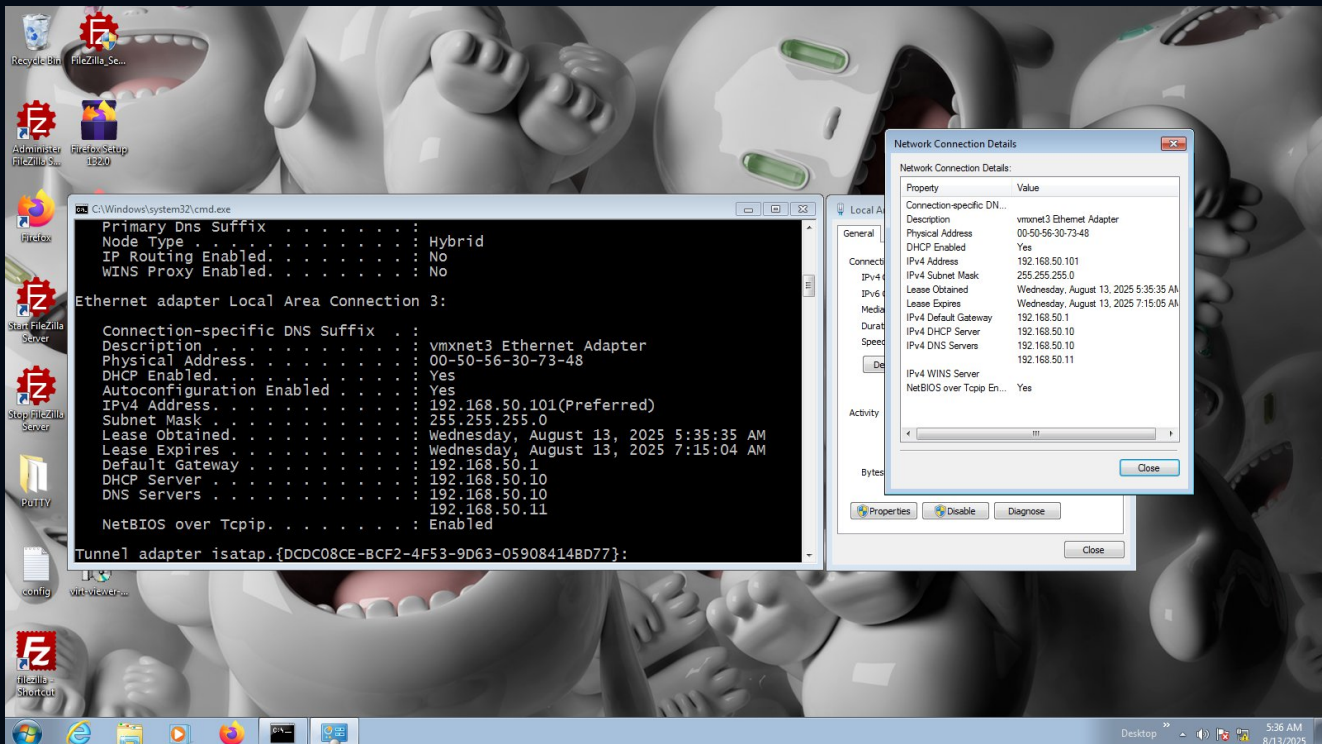
To confirm our DHCP server is working, I used a Windows 7 client:

```
ipconfig /release
```

```
ipconfig /renew
```

```
ipconfig /all
```

The client successfully received an IP address from server-alpha (192.168.50.10).



## Viewing the Server Logs

Next, I checked the KEA DHCP logs on server-alpha.

The logs showed the DHCP requests from the client — but also something interesting:

```
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo systemctl res
rescue      reset-failed restart
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo systemctl restart is
c-kea-dhcp4-server.service
hosniadmin@SERVER-ALPHA:~/Desktop$
```

```
INFO DHCP4_LEASE_OFFER [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0x4d10/47:
  Lease 192.168.50.101 will be offered
INFO DHCP4_PACKET_SEND [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0x4d662708:
  trying to send packet DHCPPOFFER (type 2) from 192.168.50.10:67 to 255.255.255.255:68 on interface
  ens33
INFO DHCP4_QUERY_LABEL received query: [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48],
  tid=0x4d662708
INFO DHCP4_PACKET_RECEIVED [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0x4d662
  708: DHCPREQUEST (type 3) received from 0.0.0.0 to 255.255.255.255 on interface ens33
INFO DHCP4_LEASE_ALLOC [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0x4d662708:
  Lease 192.168.50.101 has been allocated for 7200 seconds
INFO DHCP4_LEASE_REUSE [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0x4d662708:
  Lease 192.168.50.101 has been reused for 5644 seconds
INFO DHCP4_PACKET_SEND [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0x4d662708:
  trying to send packet DHCPACK (type 5) from 192.168.50.10:67 to 255.255.255.255:68 on interface
  ens33
INFO DHCP4_QUERY_LABEL received query: [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48],
  tid=0xfccb7f35
INFO DHCP4_PACKET_RECEIVED [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0xfccb7
  f35: DHCPINFORM (type 8) received from 192.168.50.101 to 255.255.255.255 on interface ens33
INFO DHCP4_PACKET_SEND [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0xfccb7f35:
  trying to send packet DHCPACK (type 5) from 192.168.50.10:67 to 192.168.50.101:68 on interface e
  ns33
INFO DHCP4_QUERY_LABEL received query: [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48],
  tid=0xfccb7f35
INFO DHCP4_PACKET_RECEIVED [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0xfccb7
  f35: DHCPINFORM (type 8) received from 192.168.50.101 to 255.255.255.255 on interface ens33
INFO DHCP4_PACKET_SEND [hwtype=1 00:50:56:30:73:48], cid=[01:00:50:56:30:73:48], tid=0xfccb7f35:
  trying to send packet DHCPACK (type 5) from 192.168.50.10:67 to 192.168.50.101:68 on interface e
  ns33
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
  ttp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
  ttp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
  ttp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
```

[0] 0:[tnux]\* "SERVER-ALPHA" 08:42 13-Aug-25

These warnings tell us that server-alpha cannot communicate with server-beta in the High Availability setup.

```
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
  ttp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
  ttp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
  ttp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
```

That's expected — because we haven't configured server-beta yet.

## Setting up SERVER-BETA

The easiest way to create server-beta is to clone server-alpha.

Once the clone is ready, make these changes:

- Update the static IP address in /etc/netplan/01-network-manager-all.yaml

Change it to:

addresses:

- 192.168.50.11/24

- Change the hostname

Update /etc/hosts and /etc/hostname

Then run:

```
sudo hostname -b SERVER-BETA
```

- Update the KEA DHCP configuration

Swap the server-alpha and server-beta entries in the HA section so each server sees itself as this-server-name.

- Restart the service

```
sudo systemctl restart isc-kea-dhcp4-server
```

After these changes, check the logs again on both servers — the HA warnings should disappear, and the servers should report successful communication.

```
tcp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
tcp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
WARN HA_HEARTBEAT_COMMUNICATIONS_FAILED server-alpha: failed to send heartbeat to server-beta (h
tcp://192.168.50.11:8000/): No route to host
WARN HA_COMMUNICATION_INTERRUPTED server-alpha: communication with server-beta is interrupted
INFO COMMAND_RECEIVED Received command 'ha-heartbeat'
INFO COMMAND_RECEIVED Received command 'dhcp-disable'
INFO COMMAND_RECEIVED Received command 'lease4-get-page'
INFO COMMAND_RECEIVED Received command 'dhcp-enable'
INFO COMMAND_RECEIVED Received command 'ha-heartbeat'
INFO COMMAND_RECEIVED Received command 'dhcp-disable'
INFO COMMAND_RECEIVED Received command 'lease4-get-page'
INFO COMMAND_RECEIVED Received command 'dhcp-enable'
INFO COMMAND_RECEIVED Received command 'ha-heartbeat'
INFO COMMAND_RECEIVED Received command 'dhcp-disable'
INFO COMMAND_RECEIVED Received command 'lease4-get-page'
INFO COMMAND_RECEIVED Received command 'dhcp-enable'
INFO COMMAND_RECEIVED Received command 'ha-heartbeat'
INFO COMMAND_RECEIVED Received command 'dhcp-disable'
INFO COMMAND_RECEIVED Received command 'lease4-get-page'
INFO COMMAND_RECEIVED Received command 'dhcp-enable'
INFO COMMAND_RECEIVED Received command 'ha-heartbeat'
INFO COMMAND_RECEIVED Received command 'dhcp-disable'
INFO COMMAND_RECEIVED Received command 'lease4-get-page'
INFO COMMAND_RECEIVED Received command 'dhcp-enable'
```

## EPISODE 3: SETTING UP THE DNS SERVICE (BIND9)

With DHCP ready to hand out IP addresses, it's time to give our network the ability to resolve names into IP addresses (and vice versa) using BIND9 — one of the most popular and battle-tested DNS servers.

In our setup:

- ❖ Server-Alpha will be the Primary/Master DNS.
- ❖ Server-Beta will be the Secondary/Slave DNS.
- ❖ Zone transfers will be secured with TSIG keys.

We'll create both a forward lookup zone (lab.local) and a reverse lookup zone.

### 1. Installing BIND9

On both Server-Alpha and Server-Beta:

```
sudo apt update
```

```
sudo apt install -y bind9 bind9-utils
```

- ❖ **bind9**: the DNS server software.
- ❖ **bind9-utils**: tools like **dig** and **named-checkconf** to test and troubleshoot DNS.

## 2. Configure Nameservers in Netplan

Before touching the DNS configs, let's tell our OS to use our DNS servers — so we won't forget during testing.

Edit the netplan config:

```
sudo nano /etc/netplan/01-network-manager-all.yaml
```

Example (on Server-Beta in this case):

```
...
```

```
  ens33:
```

```
    addresses:
```

```
      - 192.168.50.11/24
```

```
    nameservers:
```

```
      addresses:
```

```
        - 192.168.50.10
```

```
        - 192.168.50.11
```

```
    search:
```

```
      - lab.local
```

## Explanation:

- ❖ **nameservers.addresses**: list of DNS servers (first is master, second is slave).
- ❖ **search**: optional domain suffix, so ping host1 works without typing ping host1.lab.local.

Apply changes:

```
sudo netplan apply
```

## 3. Create and Share the TSIG Key

We'll use TSIG (Transaction SIGNature) keys to authenticate and secure zone transfers between master and slave.

On Server-Alpha:

```
sudo mkdir -p /etc/bind/keys
```

```
sudo tsig-keygen -a hmac-sha256 lab_xfer | sudo tee  
/etc/bind/keys/tsig-lab.key
```

```
sudo chown root:bind /etc/bind/keys/tsig-lab.key
```

```
sudo chmod 640 /etc/bind/keys/tsig-lab.key
```

This creates a file like:

```
key "lab_xfer" {  
    algorithm hmac-sha256;  
    secret "BASE64_STRING_GOES_HERE==";  
};
```

On Server-Beta:

Copy **the exact same file** from Server-Alpha (manually for now since SSH isn't set up).

```
sudo mkdir -p /etc/bind/keys  
sudo nano /etc/bind/keys/tsig-lab.key
```

Paste the contents, then:

```
sudo chown root:bind /etc/bind/keys/tsig-lab.key  
sudo chmod 640 /etc/bind/keys/tsig-lab.key
```

## 4. Master (Server-Alpha) Configuration

- Edit `/etc/bind/named.conf.options`:

```
options {  
    directory "/var/cache/bind";  
  
    // Only allow queries from our LAN  
    allow-query { 127.0.0.1; 192.168.50.0/24; };  
  
    // Optional forwarders for internet DNS  
    // forwarders { 1.1.1.1; 8.8.8.8; };  
  
    dnssec-validation auto;  
    listen-on { any; };  
    listen-on-v6 { any; };  
};
```

- Edit `/etc/bind/named.conf.local`:

```
// Include TSIG key  
include "/etc/bind/keys/tsig-lab.key";  
  
// Define ACL for trusted clients  
acl "trusted" { 127.0.0.1; 192.168.50.0/24; };
```

```
zone "lab.local" {
    type master;
    file "/etc/bind/db.lab.local";

    // Allow slave with TSIG key
    allow-transfer { key "lab_xfer"; 192.168.50.11; };

    // Notify slave immediately on changes
    also-notify { 192.168.50.11 key "lab_xfer"; };

    allow-query { trusted; };
    notify yes;
};
```

```
zone "50.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.50";

    // Allow slave with TSIG key
    allow-transfer { key "lab_xfer"; 192.168.50.11; };

    // Notify slave immediately on changes
    also-notify { 192.168.50.11 key "lab_xfer"; };
};
```

```
allow-query { trusted; };  
notify yes;  
};
```

- Create the forward zone file:

```
sudo nano /etc/bind/db.lab.local
```

```
$TTL 3600  
@ IN SOA ns1.lab.local. hostmaster.lab.local. (  
    2025081401 ; serial  
    3600      ; refresh  
    900       ; retry  
    604800    ; expire  
    86400 )   ; minimum  
  
    IN NS ns1.lab.local.  
    IN NS ns2.lab.local.  
ns1 IN A 192.168.50.10  
ns2 IN A 192.168.50.11  
  
; Sample hosts  
www IN A 192.168.50.2  
api IN A 192.168.50.3
```

- Create the reverse zone file:

```
sudo nano /etc/bind/db.192.168.50
```

```
$TTL 3600
```

```
@    IN SOA ns1.lab.local. hostmaster.lab.local. (  
      2025081401 ; serial  
      3600      ; refresh  
      900       ; retry  
      604800    ; expire  
      86400 )   ; minimum
```

```
      IN      NS      ns1.lab.local.
```

```
10    IN      PTR      ns1.lab.local.
```

```
11    IN      PTR      ns2.lab.local.
```

```
2     IN      PTR      www.lab.local.
```

```
3     IN      PTR      api.lab.local.
```

- Validate and start (Master):

```
sudo named-checkconf
```

```
sudo named-checkzone lab.local /etc/bind/db.lab.local
```

```
sudo named-checkzone 192.168.50 /etc/bind/db.192.168.50
```

```
sudo systemctl restart bind9
```

```
sudo systemctl status bind9 --no-pager
```

## 5. Slave (Server-Beta) Configuration

- Edit `/etc/bind/named.conf.options`:

```
options {  
    directory "/var/cache/bind";  
    allow-query { 127.0.0.1; 192.168.50.0/24; };  
    dnssec-validation auto;  
    listen-on { any; };  
    listen-on-v6 { any; };  
};
```

- Edit `/etc/bind/named.conf.local`:

```
// Include TSIG key
include "/etc/bind/keys/tsig-lab.key";

acl "trusted" { 127.0.0.1; 192.168.50.0/24; };

zone "lab.local" {
    type slave;

    masters { 192.168.50.10 key "lab_xfer"; };
    file "/var/cache/bind/db.lab.local";

    allow-query { trusted; };
};

zone "50.168.192.in-addr.arpa" {
    type slave;

    masters { 192.168.50.10 key "lab_xfer"; };
    file "/var/cache/bind/db.192.168.50";
```

```
allow-query { trusted; };  
};
```

- Validate and start (Slave):

```
sudo named-checkconf
```

```
sudo systemctl restart bind9
```

```
sudo systemctl status bind9 --no-pager
```

Within seconds, check logs on the slave — you should see the zone transfer complete.

## 6. Testing the DNS Setup

From any client (or from the servers themselves):

Check NS records:

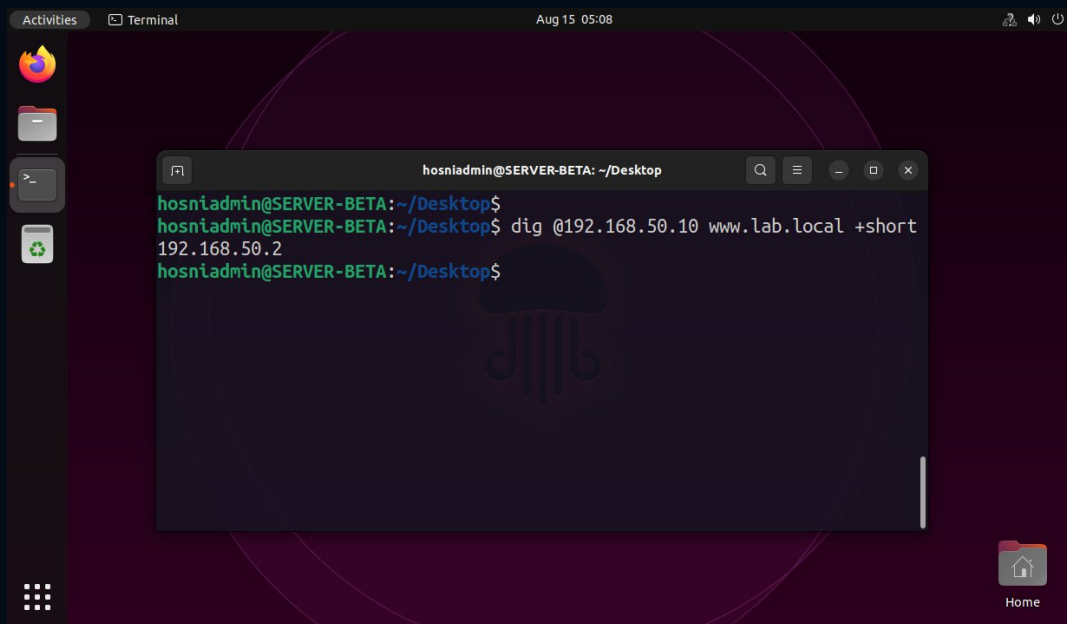
```
dig @192.168.50.10 lab.local NS +noall +answer
```

```
dig @192.168.50.11 lab.local NS +noall +answer
```

Check host resolution:

```
dig @192.168.50.10 www.lab.local +short
```

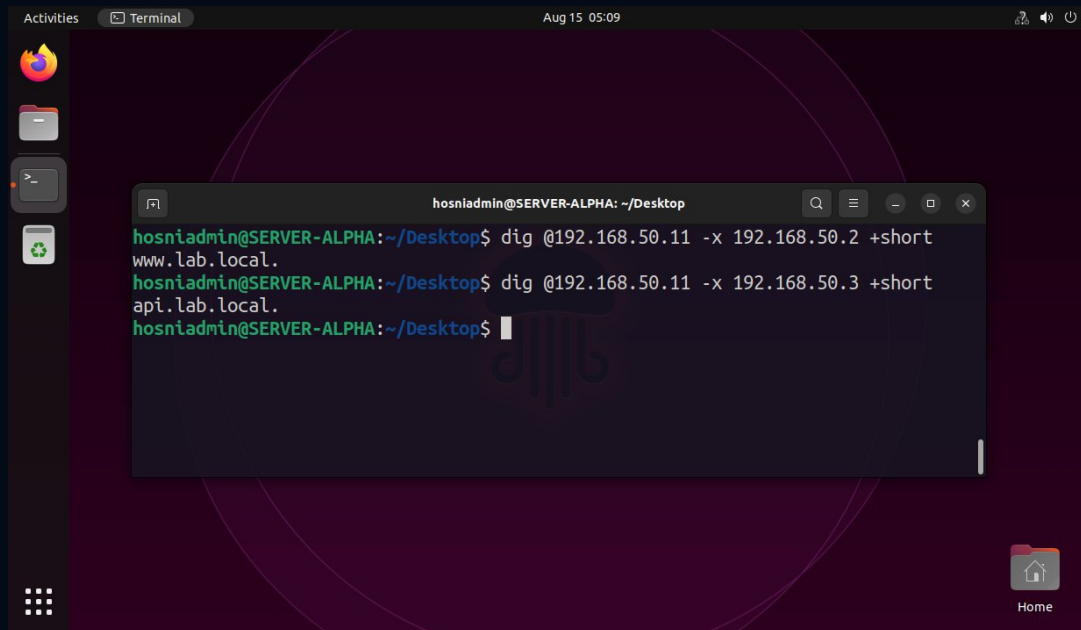
```
dig @192.168.50.11 www.lab.local +short
```



Check reverse host resolution

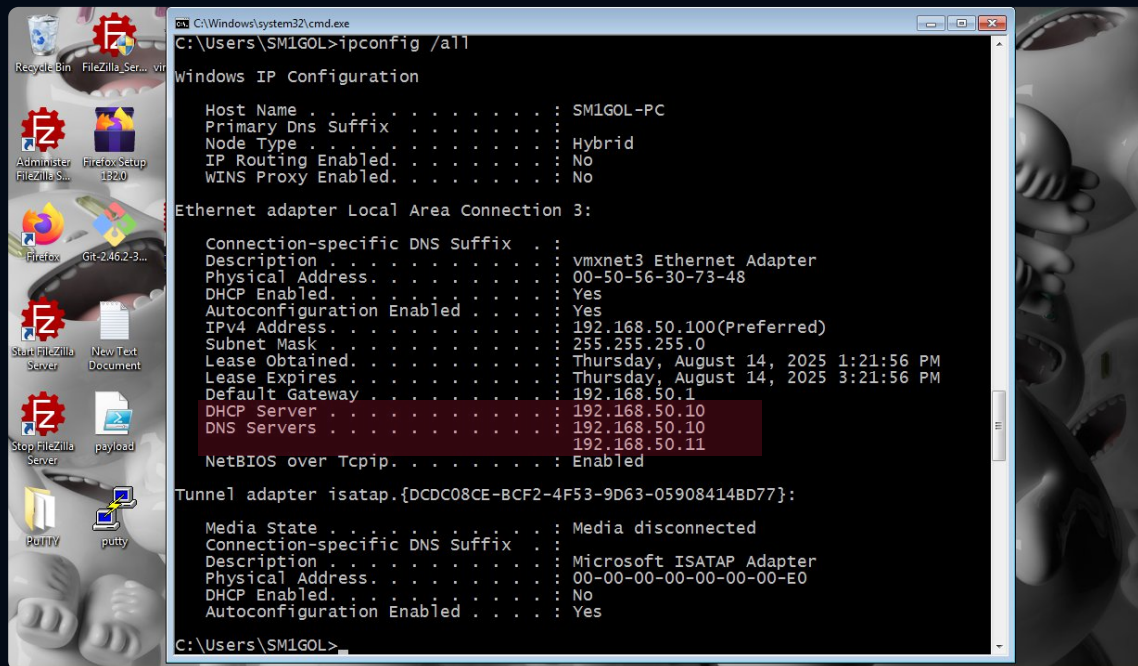
```
dig @192.168.50.10 -x 192.168.50.2 +short
```

```
dig @192.168.50.11 -x 192.168.50.2 +short
```



If both servers respond correctly, congratulations –  
your DNS HA pair is live!

# FULL TEST



```
C:\Windows\system32\cmd.exe
C:\Users\SM1GOL>ipconfig /all

windows IP Configuration

Host Name . . . . . : SM1GOL-PC
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

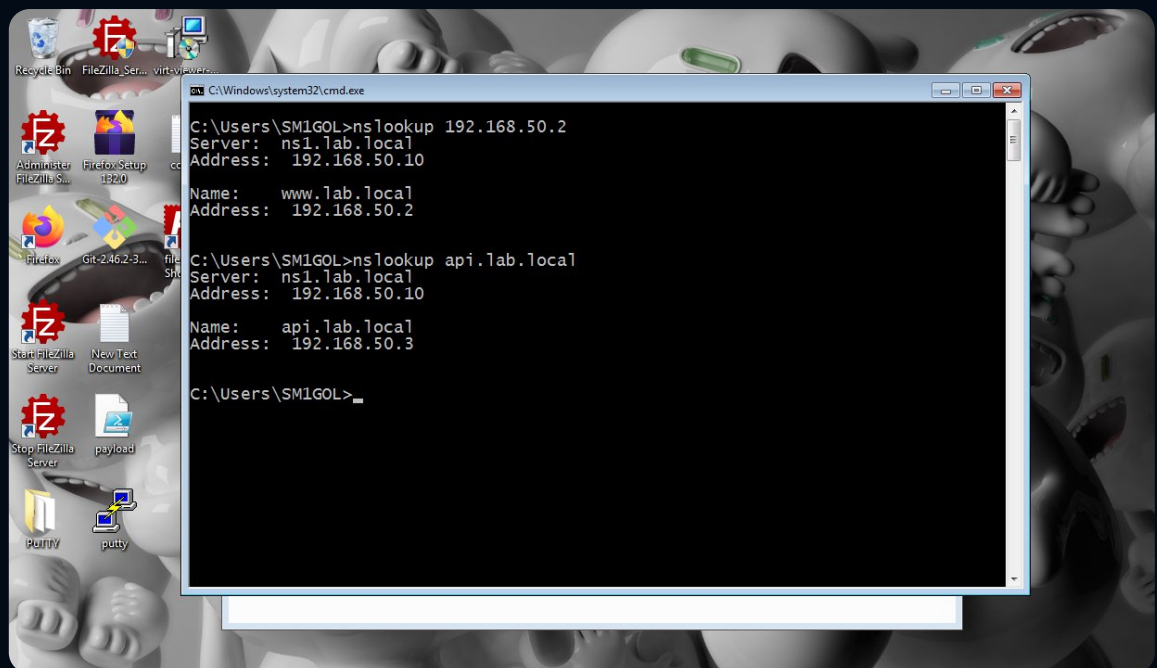
Ethernet adapter Local Area Connection 3:

Connection-specific DNS Suffix . . : vmxnet3 Ethernet Adapter
Description . . . . . : vmxnet3 Ethernet Adapter
Physical Address. . . . . : 00-50-56-30-73-48
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.50.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, August 14, 2025 1:21:56 PM
Lease Expires . . . . . : Thursday, August 14, 2025 3:21:56 PM
Default Gateway . . . . . : 192.168.50.1
DHCP Server . . . . . : 192.168.50.10
DNS Servers . . . . . : 192.168.50.10
                        192.168.50.11
NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter isatap.{DCDC08CE-BCF2-4F53-9D63-05908414BD77}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :
Description . . . . . : Microsoft ISATAP Adapter
Physical Address. . . . . : 00-00-00-00-00-00-E0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

C:\Users\SM1GOL>
```



```
C:\Windows\system32\cmd.exe
C:\Users\SM1GOL>nslookup 192.168.50.2
Server: ns1.lab.local
Address: 192.168.50.10

Name: www.lab.local
Address: 192.168.50.2

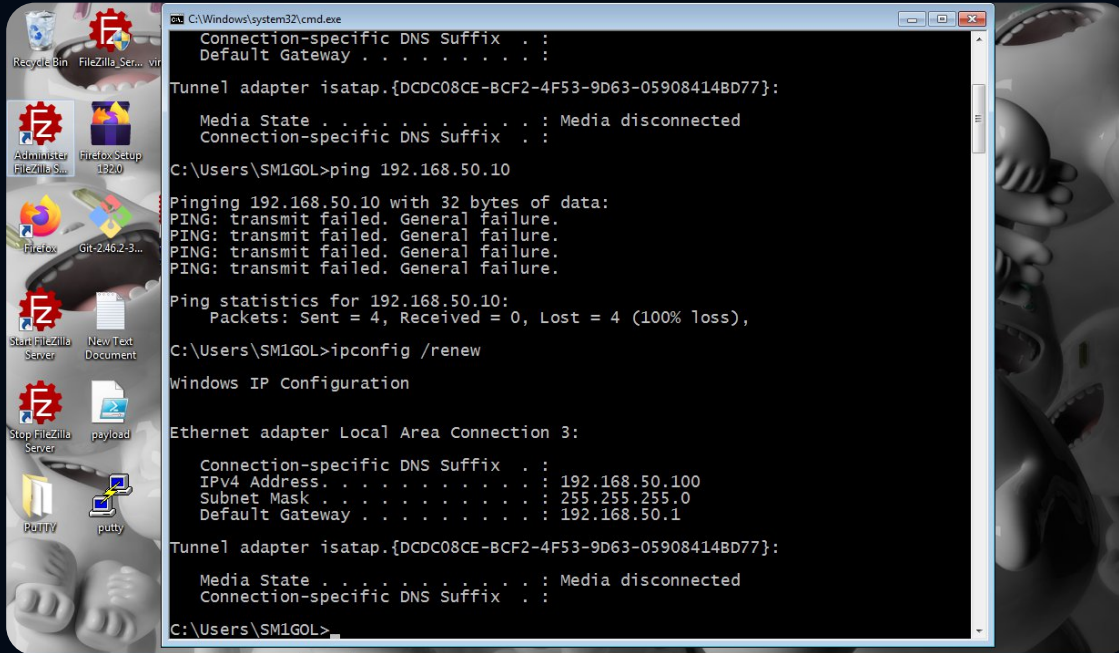
C:\Users\SM1GOL>nslookup api.lab.local
Server: ns1.lab.local
Address: 192.168.50.10

Name: api.lab.local
Address: 192.168.50.3

C:\Users\SM1GOL>
```

## Simulate failover:

- Power off SERVER-ALPHA .



```
C:\Windows\system32\cmd.exe
Connection-specific DNS Suffix . : 
Default Gateway . . . . . : 

Tunnel adapter isatap.{DCDC08CE-BCF2-4F53-9D63-05908414BD77}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\SM1GOL>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\SM1GOL>ipconfig /renew

Windows IP Configuration

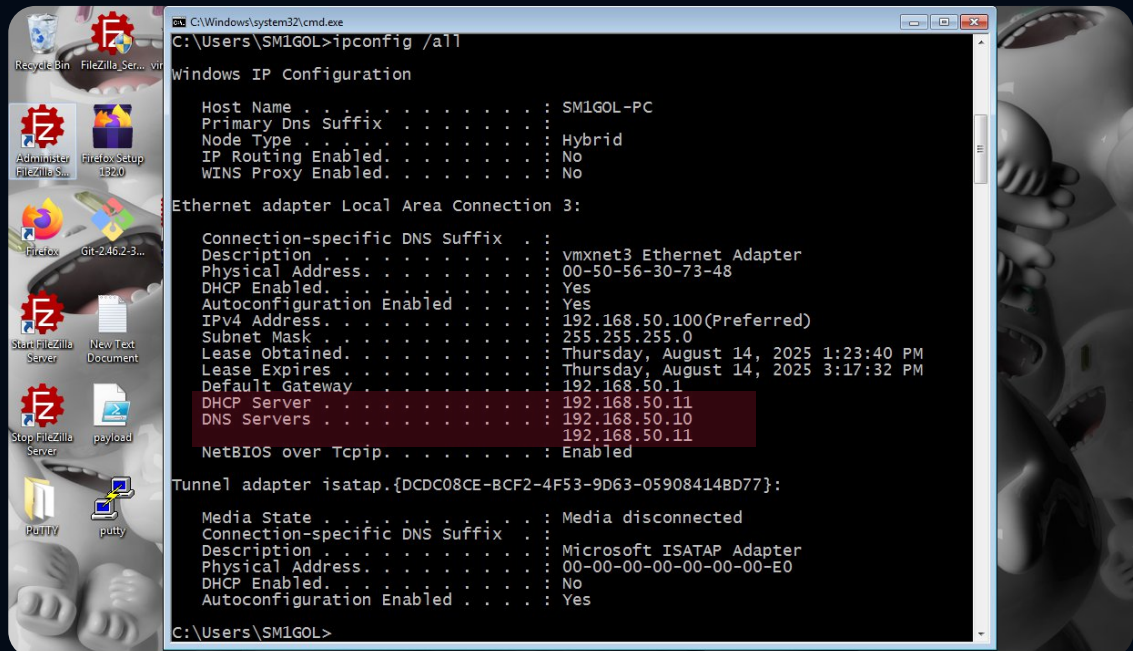
Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix . : 
    IPv4 Address. . . . . : 192.168.50.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.50.1

Tunnel adapter isatap.{DCDC08CE-BCF2-4F53-9D63-05908414BD77}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\SM1GOL>
```



```
C:\Windows\system32\cmd.exe
C:\Users\SM1GOL>ipconfig /all

Windows IP Configuration

Host Name . . . . . : SM1GOL-PC
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix . : 
    Description . . . . . : vmxnet3 Ethernet Adapter
    Physical Address. . . . . : 00-50-56-30-73-48
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . . : Yes
    IPv4 Address. . . . . : 192.168.50.100(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Thursday, August 14, 2025 1:23:40 PM
    Lease Expires . . . . . : Thursday, August 14, 2025 3:17:32 PM
    Default Gateway . . . . . : 192.168.50.1
    DHCP Server . . . . . : 192.168.50.11
    DNS Servers . . . . . : 192.168.50.10
                          192.168.50.11
    NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter isatap.{DCDC08CE-BCF2-4F53-9D63-05908414BD77}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft ISATAP Adapter
Physical Address. . . . . : 00-00-00-00-00-00-E0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . . : Yes

C:\Users\SM1GOL>
```